The Islamia University of Bahawalpur
Department of Information Technology

INFT-4041 · BSIT · 5th Semester · Fall 2026

E-Commerce Application Development

Appendix

G

The Sixteen-Week Schedule

Taught in ___

Laboratory ___

Assessment ___

This is a class extract. It is *Appendix G* of the unified handout, printed on its own so that a week's reading is a week's reading. Page numbers restart at 1 and are local to this extract; chapter, figure and table numbers are the ones the full volume gives them, so a reference you follow here lands in the same place there.

Cross-references to other chapters print as their numbers in the full volume. For the whole course in one file — front matter, the HEC tracing table, all twenty-three chapters and all seven appendices — see `ECAD_Fall2026_Handout.pdf`.

Dr. Mustafa Hameed · Department of Information Technology

Generated from `parts/` by `makechapters.py`; do not edit this PDF's source by hand

Chapter G

The Sixteen-Week Schedule

This appendix is the **source of record** for the teaching schedule. The one-page `ECAD_Fall2026_Lecture_Schedule.pdf` distributed to students is generated from it, and the slide pipeline parses the table below. Change this first; regenerate the one-page version afterwards.

Twenty-three chapters are taught across sixteen weeks, so seven weeks carry two chapters. The pairings are not arbitrary. In every case the two chapters are either one topic the printed book splits for length — the catalogue and its variants, the two halves of the back office — or a pair in which the second is unusable without the first: a basket with nothing to check out, a gateway with no payment method chosen.

One pairing is deliberate in the other direction. **Chapter 16, Security, has a week to itself**, although it would pair naturally with Chapter 15. Its laboratory attacks the student's own shop and then hardens it, and that does not fit in half a week alongside anything else.

G.1 The Schedule

Wk	Lecture topic	Chapters	Lab	Assessment
1	What e-commerce changed; business and revenue models	1, 2	Read four real shops; the arithmetic of one order	—
2	Auctions, portals and communities; planning a framework	3, 4	Three shapes compared; your requirements and your decision	—
3	The stack, and your shop in one command	5	Compose up; verify, break and reset the shop	—
4	Products, categories and the data model beneath them	6	50,000 products, and the index that rescues the category page	—
5	Product variations, options and user uploads	7	Option combinations, and what an upload field accepts	**Quiz 1 (Ch 1–5)**
6	Themes, layout and the user experience	8	Page weight before and after, measured in bytes and requests	**Assignment 1 set**
7	The shopping basket, and where state lives	9	Session, cookie and database baskets across a restart	—
8	The checkout and the order state machine	10	Midterm, then queries counted per checkout step	**Midterm (Ch 1–9)**

Wk	Lecture topic	Chapters	Lab	Assessment
9	Shipping, tax and zones; discounts and vouchers	11, 12	Geo zones and GST; a voucher applied in the wrong order	**Assignment 1 due**
10	Web payment systems; taking payment for orders	13, 14	A sandbox gateway, and what happens when it times out	**Project proposal**
11	User account management	15	Password cost against logins per second	—
12	Securing the shop	16	The OWASP checks against your own container, before and after hardening	**Quiz 2 (Ch 10–15)**
13	Administration: the dashboard, orders, customers, refunds	17, 18	A refund through the order state machine	**Assignment 2 set**
14	Extending the platform: your first module	19	A shipping method and a dashboard widget, written from nothing	—
15	Deployment, maintenance and performance; SEO	20, 21	Load test with cache on and off; crawlability measured	**Assignment 2 due**
16	Social, legal and ethical issues; project defence	22, 23	Project presentations	**Project due; presentations**

G.2 How the Assessment Adds Up

HEC publishes the assessment instruments for this course — sessional examination, home assignments, quizzes, project, presentations, final examination — without weights. The weights below are the department's, and are stated here so that a student can see the whole shape of the semester on one page.

Instrument	Weight	What it examines
Two quizzes	10%	Weeks 5 and 12. Short, closed-book, on the chapters named in the schedule
Two assignments	15%	Set in weeks 6 and 13. Both are done on your own shop and submitted as a change to it plus a short write-up
Midterm examination	20%	Week 8, Chapters 1–9
Semester project	25%	The shop itself, built across the whole semester (Chapter 23), submitted in week 16
Presentation and defence	5%	Week 16. Ten minutes, plus questions you are expected to be able to answer about your own decisions
Final examination	25%	All chapters, with emphasis after the midterm

⚠ The project is not a week-16 activity

Twenty-five per cent of the marks are for a shop that is built in the laboratory of every chapter from Chapter 5 onward. A student who treats the laboratories as optional homework and intends to build the project in the last fortnight will be building, in two weeks and without help, something the rest of the class built in fourteen weeks with a lecturer in the room.

Chapter 23 states what is required and how it is marked. Read it in Week 3, not in Week 15.